Current Group-I/II Dry-Volume Reference Model (v7.15)

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1. Status and authority

This document is the **accepted-current scientific/numerical authority** for the dry framework-compensating composition-volume model retained by v7.15. It owns:

- the supported framework normalization for the empirical volume model;
- charge-equivalent interpolation;
- the `/24` denominator;
- species-specific empirical/effective endpoint values and classifications;
- placeholder behavior;
- source-to-model transformations and their limitations;
- `volume_factor` semantics.

The chemistry, supported species, exact framework-compensating population typing, and formal-charge table are owned by `v7_13_group_i_ii_cation_extension.md`. The sparse chloride extension and the explicit exclusion of occlusion-counterion/Cl populations from this automatic volume model are owned by `v7_15_chloride_occlusion_extension.md`. Module specifications and code concretize these authorities.

2. Background and terminology

For the supported Al24Si24O96 LTA cell, let:

- $N_i = N_i^{\mathrm{fw}}$ be the exact integer number of **framework-compensating** cations of species i;
- z_i be its formal positive charge (+1 for Group I, +2 for Group II);
- $V_i^{(24)}$ be the dry-LTA **reference corner volume** assigned to species i, normalized to one 24-Al reference cell;
- $w_i = z_i N_i/24$ be the species' charge-equivalent fraction;
- f be `volume_factor`.

An endpoint may be **DIRECT**, **EXTRAPOLATED**, **PLACEHOLDER**, or **QUESTIONABLE**. These classifications describe the evidence/source-to-model status of the endpoint; they are not uncertainty distributions.

3. Supported framework scope

The model is defined only for the current production normalization:

```
Al24Si24O96
Si:Al = 1:1
sum_i z_i N_i^fw = 24
```

The empirical $V_i^{(24)}$ values are 24-Al reference-cell volumes. Non-24-Al LTA cells, other Si:Al ratios, and alternative cell normalizations are outside v7.15 scope.

A future extension must explicitly define both the endpoint normalization and the mapping from normalized composition volume to generated-cell volume. Support must not be inferred merely by replacing the fixed denominator 24 with an arbitrary N_{Al}.

4. Governing equations

For every supported neutral composition,

$$\sum_i z_i N_i = 24, \qquad w_i = \frac{z_i N_i}{24}, \qquad \sum_i w_i = 1,$$

and

$$V_{\mathrm{ref}} = \sum_i w_i V_i^{(24)} = \frac{1}{24} \sum_i z_i N_i V_i^{(24)}.$$

The final target volume is

$$V_{\mathrm{target}} = f V_{\mathrm{ref}}.$$

Isotropic lattice scaling is

$$s = \left(\frac{V_{\mathrm{target}}}{V_{\mathrm{old}}} \right)^{1/3}, \qquad A_{\mathrm{new}} = s A_{\mathrm{old}}.$$

Fractional coordinates are unchanged by this pure lattice scaling.

For Li/Na/K-only cells, all $z_i = 1$, so this expression is exactly the historical count-over-24 interpolation.

5. Source-to-model transformations

5.1 Cell normalization

For a conventional cubic Fm-3c LTA cell with lattice parameter a, the package's 24-Al reference volume is

$$V^{(24)}_{\mathrm{Fm\bar{3}c}}(a) = \frac{a^3}{4}.$$

For the 12-Al Pm-3m pseudocell convention used for the Ba source,

$$V^{(24)}_{\mathrm{Pm\bar{3}m}}(a) = 2a^3.$$

These two transformations prevent 12 Å / 24 Å LTA convention confusion.

5.2 Affine extraction of a target corner from a dry Na/M mixture

When a source provides a dry Na/M mixed composition rather than a pure endmember, the effective target corner is derived in **charge-equivalent space**. Let q_M and q_{Na} be the target and Na charge equivalents in the source normalization, with $q_{\mathrm{tot}} = q_M + q_{\mathrm{Na}}$. Then

$$V^{(24)}_M = \frac{q_{\mathrm{tot}} V^{(24)}_{\mathrm{mix}} - q_{\mathrm{Na}} V^{(24)}_{\mathrm{Na}}}{q_M}.$$

For monovalent alkalis, charge equivalents equal ion counts. For Mg and Ba, the Group-II population contributes twice its ion count.

5.3 Placeholder rule

If the package has no bound defensible species-specific dry crystallographic endpoint for a supported species, that species alone is assigned the standard Na-A endpoint:

$$V^{(24)}_i = V^{(24)}_{\mathrm{Na}}.$$

This substitution acts only on that species' own w_i contribution. A placeholder species never forces an entire mixed composition to use the Na volume.

6. Bound endpoint table

Species	Source basis	Lattice parameter used	$V_i^{(24)}$ (A^3)	Classification
H+	no species-specific endpoint currently bound; Na-A substitution	24.55930 A via Na	3703.292036	PLACEHOLDER
Li+	dry Li92Na4-A, 300 K	23.88056 A	3391.674204	EXTRAPOLATED
Na+	dry Na96-A, 300 K	24.55930 A	3703.292036	DIRECT
K+	dry K93Na3-A, 300 K	24.55825 A	3702.801747	EXTRAPOLATED
Rb+	dry Rb76Na20-A, 300 K	24.55332 A	3699.875813	EXTRAPOLATED
Cs+	dry Cs56Na40-A, 300 K	24.49934 A	3656.907061	EXTRAPOLATED

Species	Source basis	Lattice parameter used	$V_i^{(24)}$ (A^3)	Classification
Fr+	no species-specific endpoint currently bound; Na-A substitution	24.55930 A via Na	3703.292036	PLACEHOLDER
Be2+	no species-specific endpoint currently bound; Na-A substitution	24.55930 A via Na	3703.292036	PLACEHOLDER
Mg2+	dehydrated approximately Mg2Na8-A; residual water reported	24.4244(14) A	3521.221887	EXTRAPOLATED
Ca2+	fully exchanged/dehydrated Ca48-A	24.47 A	3663.042156	DIRECT
Sr2+	two dehydrated Fm-3c refinements; nonideal Sr/extra-framework-Al chemistry	24.674(1), 24.687(1) A -> package $a_{\mathrm{eff}} = 24.68$ A	3758.161808	QUESTIONABLE
Ba2+	fully dehydrated Ba3.5Na5-A, 12-Al pseudocell	12.267(7) A	3683.692333	EXTRAPOLATED
Ra2+	no species-specific endpoint currently bound; Na-A substitution	24.55930 A via Na	3703.292036	PLACEHOLDER

7. Reproducible endpoint derivations

7.1 Na

Carey's dry Na96-A conventional cell gives

$$V_{\mathrm{Na}}^{(24)} = \frac{24.55930^3}{4} = 3703.292036385714\ \mathrm{A}^3.$$

The package stores this value at binary floating-point precision.

7.2 Li, K, Rb, and Cs

The Carey measurements are conventional dry cells with 96 charge equivalents. Their 24-Al mixed-cell volumes are $a^3/4$, followed by removal of the Na contribution:

$$V_M^{(24)} = \frac{96V_{\mathrm{mix}}^{(24)} - N_{\mathrm{Na}}V_{\mathrm{Na}}^{(24)}}{N_M},$$

because these species are monovalent.

The bound source compositions are:

```
Li92Na4-A   a = 23.88056 A
K93Na3-A    a = 24.55825 A
Rb76Na20-A  a = 24.55332 A
Cs56Na40-A  a = 24.49934 A
```

This yields the Li/K/Rb/Cs endpoint values tabulated above.

7.3 Mg

For approximately Mg2Na8-A per 12 framework Al, the charge-equivalent composition is

```
Mg: 2 ions x +2 = 4 equivalents
Na: 8 ions x +1 = 8 equivalents
total           = 12 equivalents
```

The reported conventional Fm-3c lattice parameter is `24.4244(14) A`, so

$$V_{\mathrm{mix}}^{(24)} = \frac{24.4244^3}{4},$$

and

$$V_{\mathrm{Mg}}^{(24)} = \frac{12V_{\mathrm{mix}}^{(24)} - 8V_{\mathrm{Na}}^{(24)}}{4} = 3521.221887\ \mathrm{A}^3.$$

The source reports a small residual water content despite dehydration. The endpoint is therefore an extrapolated dry-volume model corner, not an unqualified perfectly anhydrous pure-Mg measurement.

7.4 Ca

For fully exchanged/dehydrated Ca-A with conventional `a=24.47 A`,

$$V_{\mathrm{Ca}}^{(24)} = \frac{24.47^3}{4} = 3663.042156\ \mathrm{A}^3.$$

No Na-removal extrapolation is applied.

7.5 Sr

Pluth and Smith report two relevant dehydrated Fm-3c lattice parameters:

```
24.674(1) A
24.687(1) A
```

The package preserves its established v7.14 Sr corner by an explicit deterministic source reduction:

$$a_{\mathrm{midpoint}} = \frac{24.674 + 24.687}{2} = 24.6805\ \mathrm{A},$$

$$a_{\mathrm{eff}} = \mathrm{round}(24.6805, 2) = 24.68\ \mathrm{A},$$

$$V_{\mathrm{Sr}}^{(24)} = \frac{24.68^3}{4} = 3758.161808\ \mathrm{A}^3.$$

`24.68 A` is therefore a **package representative**, not a single directly reported source measurement. The source chemistry is also nonideal for a pure Sr6-A interpretation: the refinements contain approximately 5.7 Sr per 12-Al pseudocell and an extra-framework Al complex. The endpoint remains explicitly QUESTIONABLE.

7.6 Ba

For fully dehydrated Ba3.5Na5-A in the 12-Al Pm-3m pseudocell,

$$V_{\mathrm{mix}}^{(24)} = 2(12.267)^3.$$

Its charge-equivalent composition is

```
Ba: 3.5 ions x +2 = 7 equivalents
Na: 5 ions  x +1 = 5 equivalents
total              = 12 equivalents
```

so

$$V_{\mathrm{Ba}}^{(24)} = \frac{12V_{\mathrm{mix}}^{(24)} - 5V_{\mathrm{Na}}^{(24)}}{7} = 3683.692333\ \mathrm{A}^3.$$

The source reports instability of fully Ba-exchanged LTA toward dehydration. This endpoint is therefore an affine model corner inferred from a dry partially exchanged composition, not a claim that a stable dry Ba6-A endmember was measured.

8. Placeholder species

For H, Be, Fr, and Ra, the package currently binds no species-specific empirical dry-LTA volume endpoint. Each therefore uses

$$V_i^{(24)} = V_{\mathrm{Na}}^{(24)} = 3703.292036385714\ \mathrm{A}^3.$$

This is a package-provenance statement, not an exhaustive assertion that suitable literature data do not exist. A future evidence review may replace a placeholder only by explicitly binding a defensible source and source-to-model transformation.

9. Scientific limitations

The dry model introduced in v7.14 and retained unchanged for framework-compensating counts in v7.15 is an affine empirical volume model for generating starting structures. It does not claim:

- excess/mixing volumes are zero in real materials;
- all endpoint measurements share identical temperature or hydration history;
- extrapolated corners correspond to stable isolable pure endmembers;
- the Sr QUESTIONABLE corner is an ideal pure-Sr measurement;
- the placeholder endpoints are physically equal to Na-A;
- random or maximum-symmetry cation positions are experimentally relaxed equilibrium positions;
- the model applies outside Al24Si24O96.

These limitations constrain physical interpretation but do not alter the defined numerical model.

10. Primary empirical sources

1. T. Carey, C. C. Tang, J. A. Hriljac, and P. A. Anderson, "Chemical Control of Thermal Expansion in Cation-Exchanged Zeolite A," *Chemistry of Materials* **26**, 1561-1566 (2014). DOI: 10.1021/cm403312q.
2. T. Carey, *Chemical control of thermal expansion in zeolites with the LTA topology*, PhD thesis, University of Birmingham (2013), Chapter 3, especially Tables 3.1 and 3.5-3.9. Stable repository record: https://etheses.bham.ac.uk/id/eprint/4258/.

3. J. M. Adams and L. V. C. Rees, "The structure of a dehydrated partially magnesium exchanged zeolite A (~Mg2Na8-A) by neutron profile refinement," *Journal of Solid State Chemistry* **62**, 184-190 (1986). DOI: 10.1016/0022-4596(86)90230-6.
4. F. Porcher, M. Souhassou, H. Graafsma, A. Puig-Molina, Y. Dusausoy, and C. Lecomte, "Refinement of framework disorder in dehydrated CaA zeolite from single-crystal synchrotron data," *Acta Crystallographica B* **56**, 766-772 (2000). DOI: 10.1107/S0108768100006893.
5. J. J. Pluth and J. V. Smith, "Crystal structure of dehydrated strontium-exchanged zeolite A. Absence of near-zero-coordinate strontium(2+) ion. Presence of aluminum complex," *Journal of the American Chemical Society* **104**, 6977-6982 (1982). DOI: 10.1021/ja00389a015.
6. Y. Kim, V. Subramanian, R. L. Firor, and K. Seff, "Four Crystal Structures of Ba_xNa_(12-2x)-A (1 <= x <= 6) Relating to the Instability of Barium-Exchanged Zeolite A Toward Dehydration," *ACS Symposium Series* **135**, 137-153 (1980). DOI: 10.1021/bk-1980-0135.ch007.

11. Historical note

v7.14 replaced the temporary v7.13 input-template-volume fallback with the current evidence-bound endpoint table and charge-equivalent interpolation. The filename is retained for release lineage and compatibility; the semantics above are the current v7.15 dry-volume authority.

v7.15 occlusion compatibility boundary

The accepted v7.14 dry-volume model is unchanged. In v7.15 its population argument is explicitly the **framework-compensating** role only:

```
V_ref = (1/24) sum_i z_i N_i^fw V_i^(24)
```

Neither `occlusion_counterion` cations nor Cl cage guests participate in the automatic interpolation. This preserves the qualified dry reference as an initialization policy; it is not evidence that salt loading causes zero physical volume change.